Real Name, Real Face, Real Talk? Anonymity and Toxicity on Mastodon

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Abstract

This study investigates the relationship between user anonymity and toxicity in discourse on Mastodon - a decentralized social media platform. We develop a profile-based anonymity classification framework using name and face presence as identity cues, and apply a pre-trained language model to detect toxicity in user-generated posts. Our results show that identifiable users, particularly those who disclose both name and face, tend to post less toxic content. We also find that higher levels of instance moderation are associated with reduced toxicity overall. These findings highlight how identity presentation and platform governance jointly shape discourse quality in decentralized social networks.

Introduction

Decentralized social media platforms such as Mastodon have emerged as privacy-aware, community-driven alternatives to centralized networks like Twitter or Facebook. Their federated architecture distributes control over data, moderation, and policy enforcement across independently operated servers called *instances* (Zignani, Gaito, and Rossi 2018; Raman et al. 2019). While this infrastructure empowers users and administrators, it also introduces challenges for maintaining consistent moderation practices and understanding user behavior across heterogeneous communities.

A key behavioral factor in online environments is *anonymity*. Anonymity can encourage participation by lowering social inhibition and enabling users to express themselves freely, particularly on sensitive or controversial topics (Kim et al. 2023). However, it may also lower accountability, increasing the likelihood of toxic behaviors such as harassment or verbal aggression, an effect known as online disinhibition (Suler 2004). These dynamics have been examined on centralized platforms, but their relevance and strength in decentralized settings, where identity policies and community norms vary, remain underexplored.

Toxicity, defined here as hostile, offensive, or aggressive language, poses a threat to the quality and inclusiveness of online discourse. Although Mastodon instances typically define rules prohibiting hate speech, threats, and harassment, the degree of enforcement varies substantially. Moreover, Mastodon's relaxed stance on identity verification allows for a diversity of anonymity, ranging from pseudonymous usernames to full real-name profiles with profile pictures. This makes Mastodon an ideal research platform for studying how different forms of anonymity relate to toxic behavior in loosely governed social environments.

In this study, we examine how user anonymity is associated with toxic content production on Mastodon. We focus on how different forms of anonymity, based on name and face presence, relate to user-level toxicity across a diverse set of federated communities. Our approach leverages large-scale behavioral data and machine learning-based toxicity detection to examine subtle variations in discourse outcomes across identity types. This work contributes to the growing literature on decentralized platform dynamics and offers implications for federated community governance.

Related Work

Decentralized Platforms and Moderation. Research on decentralized social networks (DOSNs) is still nascent. Studies have mostly focused on technical aspects such as network structure and federation dynamics (Zignani, Gaito, and Rossi 2018; La Cava, Greco, and Tagarelli 2021), or the challenges of decentralized moderation and content filtering (Raman et al. 2019). Social behavior in federated spaces remains underexplored. Mastodon's heterogeneous moderation practices and flexible identity norms offer a unique opportunity to test how platform structure mediates well-established relationships between anonymity and discourse quality.

Anonymity and Online Behavior. Anonymity has long been recognized as a key factor influencing user behavior in digital communication. Suler (2004) identifies psychological mechanisms, such as dissociative anonymity and invisibility, that explain why users behave more openly, and sometimes more aggressively, when their identities are obscured. Empirical studies show that anonymous users are significantly more likely to post uncivil or aggressive content. For example, Santana (2014) found that over half of anonymous comments on news websites were uncivil, compared to under 30% of those from identifiable users. Similarly, analysis of online communities like Reddit has shown that users who later get banned often begin with relatively mild behavior

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that escalates over time (Cheng, Danescu-Niculescu-Mizil, and Leskovec 2015).

Toxicity Definitions and Detection. Toxicity includes behaviors such as insults, threats, hate speech, and offensive language. Detection approaches range from manual annotation (Halpern and Gibbs 2013) to supervised learning using transformer models like BERT and HateBERT (Duchêne et al. 2023). These models perform well on benchmarks but often misclassify nuanced cases, such as sarcasm or coded language (Sheth, Shalin, and Kursuncu 2022; Davidson et al. 2017). In decentralized settings, challenges such as instance-level variation and the lack of centralized moderation further complicate detection. For example, recent work explores model sharing across instances for toxicity detection (Bin Zia et al. 2022) and conversation-based moderation for the Fediverse (Agarwal et al. 2024). These findings emphasize the need for caution when interpreting classifier output, particularly in low-toxicity environments like Mastodon.

Methods

Data

We collected user- and post-level data from ten Mastodon instances selected to reflect a range of user base sizes, topical orientations, and moderation practices. These include large, general-purpose servers such as `mastodon.social` and `mastodon.world`, as well as smaller, niche ones like `freeradical.zone` and `wetdry.world`. This diversity allows us to analyze how different platform characteristics, particularly moderation practices, may relate to the quality of user discourse.

To capture the transparency of moderation practices, we developed an ordinal measure - **moderation detail**. This score, ranging from 0 to 2, reflects the degree to which moderation mechanisms are visible and structured. Specifically, we evaluate whether moderators are publicly listed, whether users can report misconduct, and whether enforcement procedures are described in a clear and accessible way. Instances that provide little or no information about moderation were scored as 0 (low detail), while those with fully documented processes and visible moderators were scored as 2 (high detail). Table 1 summarizes the selected instances with respect to their user base size, topical focus, and moderation detail score.

From these instances, we sampled a total of 6,093 users using the Mastodon API. The number of users sampled per instance is also reported in Table 1. For each user, we collected profile metadata including their display name, profile picture URL, follower and following counts, account creation date, and number of posts. For each user, we retrieved their most recent publicly available English posts, resulting in a dataset of 459,502 posts.

Anonymity Classification

To study the role of anonymity in shaping discourse quality, we develop a machine learning–based approach to identify the presence of identity cues in user profiles by user's display name and profile picture. Based on which identity cues are present, users can be grouped into different anonymity categories, allowing us to analyze variation in behavior across levels of identifiable information. We refer to these cues as name presence and image presence, respectively, to reflect that a detected name or image may not belong to the account holder. For example, a user may use a public figure's name or a generic photo. While such cases do not necessarily reveal true identity (see the **Discussion**), they still reduce perceived anonymity by signaling the presence of an identity cue. Thus, our approach focuses on the presence of such cues as a proxy for reduced anonymity:

- Name presence: whether a user's display name contains a string that resembles a real personal name.
- Face presence: whether the user's profile picture includes a recognizable human face.

Name Presence Detection. We use a pre-trained named entity recognition (NER) model based on BERT, which has been fine-tuned for identifying person names. This model has demonstrated strong performance across multilingual contexts and is applied here to detect real-name-like entities in user display names (Devlin et al. 2019; Wolf et al. 2020). The model classifies entities into categories such as person, organization, and location. For each display name, we extract tokens labeled as person entities. To reduce false positives, we further filter out single-name matches and apply regular expressions to exclude non-name tokens or usernames. The output is a binary label indicating whether the user's display name includes a real-name-like string.

Face Presence Detection. We use a face detection pipeline based on deep convolutional neural networks, specifically a ResNet-based face encoding model (Geitgey 2017). This model has achieved high accuracy on benchmark datasets such as Labeled Faces in the Wild (LFW), making it suitable for real-world identity cue detection. It classifies whether a human face is detected in the image or not. While it performs reliably on portrait images, it may also detect face-like features in AI-generated avatars or photographs of others. These are acknowledged as limitations and discussed in the **Discussion** section.

Toxicity Detection

Next, we measure the content toxicity of user-generated posts. In this study, we define "toxicity" as the presence of language that is insulting, threatening, harassing, or excessively rude in a way that is likely to make people leave a discussion. This definition aligns with prior work on toxic behavior in online communities (Davidson et al. 2017; Sheth, Shalin, and Kursuncu 2022), which emphasizes intent to harm or disrupt civil discourse as a core aspect of toxicity.

We apply a pre-trained BERT-based text classification model, fine-tuned on large-scale online comment datasets for toxicity detection. The model outputs a probability score across multiple toxicity-related categories, including general toxicity, threats, insults, and identity-based hate. For each post, we assign a toxic label if any of the toxicity probabilities exceed a standard decision threshold of 0.5. Posts that do not meet this threshold are labeled as non-toxic.

Table 1: Summary of the selected Mastodon instances.

Instance	User Count	Topical Focus	Moderation Detail	Sampled Users
mastodon.social	2,241,612	General-purpose	2	1,372
mastodon.world	186,896	General-purpose	2	1,036
fosstodon.org	62,238	Free, Open Source Software	2	1,646
c.im	67,253	General-purpose	2	421
techhub.social	81,676	Technology	1	364
aus.social	25,456	Australia / General	1	633
mstdn.party	51,360	General-purpose	0	269
noc.social	13,396	Technology	0	93
freeradical.zone	2,835	Privacy / Leftward Politics	2	75
wetdry.world	3,139	Gaming	0	184

After applying the classifier to all posts, we calculate a **toxic proportion** for each user. This measure is defined as the percentage of a user's posts that are labeled as toxic, computed by dividing the number of toxic posts by the total number of posts analyzed. This user-level score serves as a proxy for overall toxic behavior and is used in the regression analysis to assess the relationship between anonymity and discourse quality.

Regression Models for Toxicity Analysis

We estimate a series of regression models with the proportion of toxic posts per user (i.e. toxic proportion) as the dependent variable. The key independent variables are binary indicators for name presence and face presence, as well as their interaction. The reference group consists of users lacking both cues (i.e., no name and no face), allowing comparisons across anonymity levels.

To examine whether anonymity effects vary by moderation context, we interact identity cues with the instance-level moderation detail score (0–2), which reflects the clarity and perceived enforceability of moderation practices. This allows us to test whether more transparent moderation moderates the anonymity–toxicity relationship.

Control variables include instance fixed effects, post frequency, follower/following counts, account age, and average post length. Continuous predictors are log-transformed, and robust standard errors are used to account for heteroskedasticity. All predictors have VIFs below 10, indicating no serious multicollinearity concerns.

Results

Anonymity Groups and Validation

Based on the presence of identity cues in user profiles, we grouped users into four anonymity categories:

- **Name + Face**: both a real-name-like display name and a profile image containing a human face;
- **Name Only**: a real name present, but no human face;
- **Face Only**: a human face present, but no real name;
- **No Name, No Face**: neither a real name nor a human face detected.

Applying our detection models to the full dataset of 6,093 users, we found that 1,769 users (29.0%) fell into the **Name + Face** group, 986 (16.2%) into **Name Only**, 1,069 (17.5%) into **Face Only**, and 2,254 (37.0%) into **No Name, No Face**. These results indicate a wide range of identity disclosure practices on Mastodon.

To validate the reliability of our model-based anonymity detection, we conducted a manual validation of both name and face presence using a random sample of 500 user profiles. Each profile's display name was manually assessed for containing a real name. In Table 2(a), the model achieved a strong performance: accuracy = 0.954, precision = 0.935, recall = 0.964, and F_1 = 0.949, indicating high agreement with human judgment. Similarly, face presence was manually validated across 500 profile pictures. As shown in Table 2(b), the classifier also performed reliably, with accuracy = 0.886, precision = 0.886, recall = 0.859, and F_1 = 0.872.

Table 2: Confusion matrices from manual validation of anonymity and toxicity detection.

(a) Name Presence

	Prediction	
Label	Name Present	Name Absent
Name Present	216	8
Name Absent	15	261

(b) Face Presence

	Prediction	
Label	Face Present	Face Absent
Face Present	195	32
Face Absent	25	248

(c) Toxic Content

	Prediction	
Label	Toxic	Not Toxic
Toxic	72	5
Not Toxic	56	367

Toxicity Detection

To evaluate the performance of the toxicity detection model, we manually validated a random sample of 500 posts. As shown in Table 2(c), the model achieved an accuracy of 0.878, correctly identifying 439 out of 500 posts. It showed a high recall for toxic content (0.935), detecting most of the toxic posts. However, its precision was lower (0.563), indicating a tendency to over-predict toxicity. This trade-off is captured in the class-specific F_1 score of 0.705 for toxic posts and 0.924 for non-toxic posts. To account for the imbalance between toxic and non-toxic posts in the dataset, we also report a weighted F_1 score of 0.890, which provides a more representative summary of overall model performance under natural class distribution.

Figure 1 shows how the mean proportion of toxic posts per user varies across anonymity groups and levels of moderation detail on Mastodon. We observe a clear pattern: users who are more anonymous, particularly those in the **No Name, No Face** group, tend to post a higher proportion of toxic content, especially when situated in the instances with low moderation detail. For example, the **No Name, No Face** users on low-moderation instances have the highest average toxicity (0.087), while the **Name + Face** group maintains the lowest toxicity across all moderation levels, dropping to 0.022 in highly moderated instances. These findings suggest that both identity cues and detailed moderation policies are associated with lower average toxicity per user, indicating their potential joint role in supporting healthier discourse.

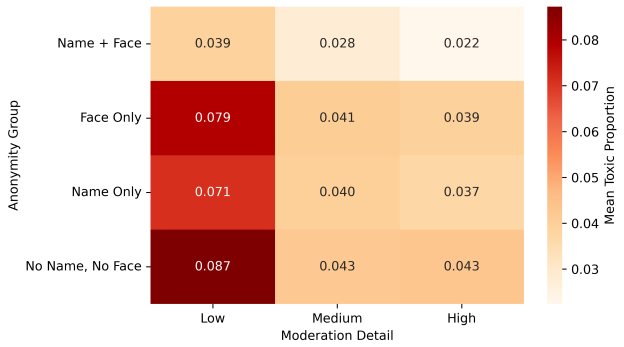


Figure 1: Mean toxic proportion by anonymity group and moderation detail.

Regression: Anonymity, Moderation, and Toxicity

Table 3 presents the regression results demonstrating the relationship between anonymity and toxicity in posting behaviors. Columns (1) to (3) show the main effects of name presence, face presence, and both (name and face) presence on toxicity, using alternative model specifications. As we can see, both name presence and face presence contribute to reduced content toxicity on Mastodon. Having an identifiable name has a larger negative effect on toxicity than having a face in the profile picture. After controlling for instance fixed effects, users with both name and face present have significantly lower levels of posting toxicity than other users.

We further analyzed how the moderation details of different instances interact with the effects of name and face presence. As shown in column (4) in Table 3, in instances with low moderation level, users with only face presence tend to post more toxic content, while users with both name and face present tend to post less toxic content. In contrast, in instances with higher moderation levels, users with only face presence tend to post less toxic content. These results indicate that instance-level moderation is associated with reduced toxicity, especially for users with face presence. Overall, even without strict instance moderation, users with face and name present tend to post less toxic content.

Table 3: Regression Results

	(1)	(2)	(3)	(4)
Name Present	-0.010***	-0.007***	-0.003	0.005
	(0.003)	(0.003)	(0.003)	(0.008)
Name Present × Moderation Detail				-0.005
				(0.005)
Face Present	-0.006**	-0.004	-0.000	0.014**
	(0.003)	(0.003)	(0.003)	(0.007)
Face Present × Moderation Detail				-0.009*
				(0.004)
Name Present × Face Present	-0.009**	-0.008**	-0.010***	-0.026**
	(0.004)	(0.004)	(0.004)	(0.011)
Name Present × Face Present × Moderation Detail				0.009
				(0.006)
Constant	0.049***	0.173***	0.128***	0.126***
	(0.002)	(0.028)	(0.031)	(0.031)
Instance FE	N	N	Y	Y
Control Variables	N	Y	Y	Y
N	6093	6093	6093	6093
adj. R^2	0.021	0.060	0.101	0.101

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Discussion

Our findings provide new evidence on how anonymity shapes toxicity in user-generated content on Mastodon. The regression analysis reveals that users who include identifiable information, such as a name or a face in their profile, tend to post less toxic content. Importantly, having a real name tends to reduce more content toxicity than simply showing a face in a user's profile. This suggests that text-based identity cues, such as names, may carry greater reputational weight or social accountability than visual cues alone. Moreover, users who disclose both name and face exhibit

the lowest levels of toxicity, suggesting a compounding effect of identity cues on promoting prosocial behavior. These patterns align with psychological theories on accountability and self-presentation in online contexts (Kim et al. 2023).

The role of platform governance also serves as an important moderating factor. We find that in instances with low levels of moderation, users with only face present are more likely to post toxic content, whereas users with both name and face present are less likely to do so. In contrast, in instances with higher levels of moderation, users with face presence tend to post less toxic content regardless of whether their name is disclosed. This suggests that moderation efforts can mitigate the risks associated with partial identity disclosure, especially for users who might otherwise feel emboldened by visual anonymity. Together, these results emphasize the interaction between platform design and governance structures in shaping discourse quality.

While our study reveals clear patterns, it is not without limitations. First, our analysis relies on automated models to detect anonymity and toxicity. Although we manually validated a subset of predictions, future work could incorporate more comprehensive human annotation to improve label reliability, especially for edge cases or culturally nuanced expressions. In addition, future work may consider fine-tuning these models, particularly the toxicity detection model, which despite achieving high recall demonstrates only moderate precision and may overflag non-toxic content as toxic. Second, our definition of anonymity is based on the presence of name- and face-like identity cues in user profiles, rather than verified identity disclosure. However, users may employ names or images that are not their own, leading to potential misclassification. Some users may also signal identity in other ways (e.g., linking to personal websites or including location tags) that are not captured by our current approach. Furthermore, the binary classification of anonymity is likely to oversimplify the nuances of identity disclosure and perceived identifiability. Therefore, a more nuanced understanding and representation of anonymity would benefit from richer profiling and possibly user survey data.

Conclusion

This study demonstrates that identity cues such as name and face presence are associated with lower levels of toxicity on Mastodon, and that these effects are shaped by instance-level moderation practices. The results suggest that both individual choices about identity presentation and platform governance contribute to healthier discourse. This work adds to ongoing conversations about anonymity and content moderation in decentralized social media platforms, offering empirical evidence on how identity presentation interacts with toxicity on such platforms.

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Paper Checklist

1. For most authors...
   (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? Yes,there is no violation of social contracts. It uses publicly available data from Mastodon, respects user anonymity, and avoids profiling or targeting individuals or groups. The analysis focuses on aggregated patterns and does not profile or infer sensitive data.
   (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope? Yes, the abstract and introduction sections summarize the research goals, methods, findings and contributions.
   (c) Do you clarify how the proposed methodological approach is appropriate for the claims made? Yes, see the Methods.
   (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? Yes, potential artifacts include uneven representation across instances and users, cultural variations in identity presentation and language use.
   (e) Did you describe the limitations of your work? Yes, see the Discussion section.
   (f) Did you discuss any potential negative societal impacts of your work? No, we do not anticipate negative societal impacts from this work. The study uses public data and focuses on aggregate patterns. It does not involve sensitive personal information or profiling, and aims to inform healthier online discourse.
   (g) Did you discuss any potential misuse of your work? No, we did not explicitly discuss potential misuse. However, we note that the methods are designed for aggregate analysis and not for identifying or targeting individuals. The study does not provide tools or guidance that would enable misuse.
   (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? No, we did not formally document mitigation steps, as this is a work-in-progress study with limited space, and no data or models are being released at this stage. However, the analysis is conducted at the aggregate level to minimize risk, and no sensitive personal information is used or shared. We plan to document and release resources responsibly in the final version.
   (i) Have you read the ethics review guidelines and ensured that your paper conforms to them? Yes, we have read the ethics review guidelines and ensured that this paper conforms to them.
2. Additionally, if your study involves hypotheses testing...
   (a) Did you clearly state the assumptions underlying all theoretical results? NA
   (b) Have you provided justifications for all theoretical results? NA
   (c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? NA
   (d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? NA
   (e) Did you address potential biases or limitations in your theoretical framework? NA
   (f) Have you related your theoretical results to the existing literature in social science? NA
   (g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? NA
3. Additionally, if you are including theoretical proofs...
   (a) Did you state the full set of assumptions of all theoretical results? NA
   (b) Did you include complete proofs of all theoretical results? NA
4. Additionally, if you ran machine learning experiments...
   (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? No, this is a work-in-progress paper, so we have not yet released the code, data, or reproduction instructions. However, the materials are available upon request, and we plan to make them publicly accessible in the final version.
   (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? NA. Since we use pre-trained models without additional training or fine-tuning, there are no training details to report. Our analysis relies on applying these models for inference only.
   (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? NA. We did not report error bars, as our work uses pre-trained models for inference without stochastic training or repeated runs. Results are deterministic given fixed inputs.
   (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? No, this work does not utilize high-performance computing or cloud providers, so these details are not included.
   (e) Do you justify how the proposed evaluation is sufficient and appropriate to the claims made? Yes, we justify our evaluation by manually validating model predictions for both anonymity and toxicity detection on a labeled sample. This validation supports the reliability of our automated annotations and ensures that the evaluation aligns with the study's claims about behavioral patterns.

(f) Do you discuss what is "the cost" of misclassification and fault (in)tolerance? Yes, we discuss model precision, recall, and F_1 scores, particularly noting the trade-off between false positives and false negatives in toxicity detection.

5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...
   (a) If your work uses existing assets, did you cite the creators? NA
   (b) Did you mention the license of the assets? NA
   (c) Did you include any new assets in the supplemental material or as a URL? NA
   (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? NA
   (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? NA
   (f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see **?**)? NA
   (g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see **?**)? NA
6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...
   (a) Did you include the full text of instructions given to participants and screenshots? NA
   (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? NA
   (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? NA
   (d) Did you discuss how data is stored, shared, and deidentified? NA